AI v.s Humans: Should AI Play A Greater Role Than Humans In Cardiovascular Procedures?

Author: Kameron Chambers
Grade: 12th Grade [Senior In High School], Independent Study

Abstract

Artificial intelligence (AI) had become increasingly integrated into modern day society. It has especially had a growing presence within the medical field. This study aims to investigate whether or not an AI model trained to read electrocardiograms (ECGs) can perform with greater accuracy and speed than humans. By using a dataset of ECG recordings both AI and human clinicians analyzed cardiac patterns in detection for potential abnormalities. The results of this study indicate that AI was able to demonstrate a higher accuracy in ECG readings than humans but fell back on the speed aspect of it. These findings highlight the potential benefits AI could pose for the medical field and industries within it. These findings also highlight AIs potential ability to enhance the reliability of cardiovascular diagnostics and with enhancements to the program, it has a nearly perfect chance at enhancing the efficiency of cardiovascular diagnostics. While this study does promote the idea of future research studies it also forms crucial considerations necessary to consider before implementing AI into such a crucial and life-dependant industry.

Introduction

AI is no longer a thought nor a concept of the future, it's something that has been deeply integrated into modern day society. Many use AI to answer complex questions while others use it to bring their creations to life. AI has slowly but surely incorporated itself into many fields of employment. As AI continues to evolve, many debate the extent to which AI will be used in the work force, especially the medical field (as it currently has a growing presence within it). People also debate the risks that AI could bring to patients as it becomes incorporated into medical facilities; wondering what the future has planned for AI.

This is a common yet complex system many people reach out to medical professionals for assistance with: The cardiovascular system. Crucial towards keeping the everyday human living. Mistakes & lengthy durations of time can have severe impacts on patients encountering complications with their cardiovascular system. More common than not, many heart based complications are first discovered through ECG readings. With this being understood, the research question being asked today is: "Does an AI model, trained to detect cardiac abnormalities in a ECG, perform with greater accuracy and speed than a human trained to read ECGs?"

The importance of this question serves a grand purpose and could help towards determining the limitations, if any, of AI being used in the medical field. Estimating the pros and cons of technical devices being implemented into industries that are crucial towards the well-being of humans, is something that needs to continue.

Important Terms & Phrases: *AI, Cardiovascular System, Integrated/-ation, ECG (Electrocardiogram), Cardiac Abnormalities, Work Force, Medical Facilities, Lengthy Duration, Future (10+), Technical Devices.*

Background

What is AI? AI stands for artificial intelligence and it's a complex system of codes created by humans that executes tasks typically requiring human intelligence. Through these sets of rules and codes, AI was made to solve complex problems, operate intelligently (providing the best answers or results) and adapt to many different types of environments depending on the scenario. AI was first introduced in 1950 by Alan Turing who created and published the idea of "*Computer Machinery and Intelligence*." AI has since then improved greatly with the most novel and public forms of AI being ChatGPT and DALL-E.

AI was first introduced into the medical field in the 1970s when "MYCIN", a strategic tool used to identify bacteria causing blood based infections, was created. As AI continued to grow, an association abbreviated AAAI (The Association for the Advancement of Artificial Intelligence) was implemented in 1979. Between the years of 1980 and 1990, AI in the medical field started advancing greatly, impacting the lives of patients positively. AI has been a tool in the medical field that has allowed for proficient data collection and more advanced research opportunities. As of 2025, AI is currently being used in many types of medical machinery such as x-rays and it's even applied to patient care with concepts such as forming diagnosis as well as treatments for patients.

The cardiovascular system however, is a complex system consisting of the heart and blood vessels; designed to use the blood running throughout the body as a way to insert what is needed (such as O_2) and remove what is not (such as CO_2). This system is more than necessary to keep humans alive and healthy. As of April, 2025, many people see AI as a transformative that could improve the speed and accuracy of healthcare; as expressed in other research studies through PubMed [NIH]. Improving the speed and accuracy of cardiovascular directed procedures, is exactly what is need in order to improve patient care within medical facilities. The relativity of this study today, not only allows for further consideration of AI having a role in the medical field, but it also poses more questions to be studied and answered.

Important Terms & Phrases: *Artificial Intelligence, Code(s), Human Intelligence, Complex Problems, Operate(-tion), Adaptation (Adapt), Novel, Blood Infection(s), X-Ray, AAAI, Proficient, Advanced Research Opportunities, Diagnosis, Blood Vessels.*

Methodology

In this research study, both the works of humans and AI were incorporated in order to determine the speed and accuracy of AI versus humans in ECG readings. In order to perform this task and collect data, a set of literary and public research studies were gathered to determine the human portion of this study. Secondly, a month of programming and training AI to analyze ECGs in a terminal program was executed. Once the program successfully ran, a public database (MIT-BIH Arrhythmia Database) that consisted of ECG readings was accessed. 15 of those ECG readings were provided into the coded AI terminal (three times) for three trials to not only test speed and accuracy but also eliminate the risk of AI guessing. Then, a timer was set to see how long it takes for the AI reader to produce an answer. The AIs answer was compared to the correct answer listed in the database. The criteria focused on in this study was **(1)** accuracy - how accurate were the answers of the humans v.s the answers generated by AI? Followed by, **(2)** speed - how quickly were both the human and AI able to provide an answer? Finally, literary information from public databases such as NIH, PubMed and MayoClinic were examined in order to collect arguments, definitions and opinions regarding this topic of research. These sources were selected based off of privacy statements, amount of access towards the information provided and how credible the source is. The decision to stay with well-known sources that were also certified for the public was used towards this research.

STEPS OF RESEARCH (In days)

- **Day 1 - Day 4**: Find an interesting topic to research.
- **Day 5 - Day 11**: Create an outline and to-do list that will execute organization throughout the duration of the research.
- **Day 12 - Day 20**: Find a credible database consisting of mock ECGs, find research articles with data fit for this study regarding the human portion of this study.
- **Day 21 - Day 40**: Create and train an AI model to read inputted ECGs
- **Day 41 - Day 48**: Review, test and fix any bugs with the AI ECG reader
- **Day 48**: Have AI read the 15 ECGSs provided and collect AI's data
- **Day 49 - Day 56**: Start writing research paper and citing all external sources
- **Day 57 - Day 60**: Create graphs and visually input all data
- **Day 60 - Day 67**: Review all work, check grammar and check data
- **Day 68 - Day 70:** Fill out forms for the Regeneron Science Talent Search

Findings

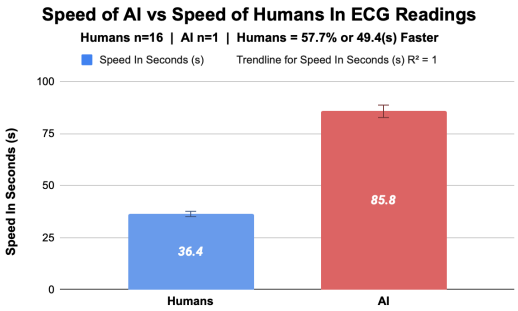
In this research study, the data collected was used towards determining two things: (1) Who produces answers regarding whether or not an ECG is normal or abnormal at a quicker rate, AI or humans? (2) Is AI or are humans more accurate in the answers they tend to express?

It was found in this study that humans tend to be quicker at producing an answer in comparison to AI. The following information was collected from a publicly available research study conducted by Tahri Sqalli et al. (2021), which used 16 medical students in order to analyze 12-lead ECGs and collect data regarding speed, accuracy and eye-movement/eye-tracking. In this study, it was determined that on average it takes between 13.2 seconds and 59.5 seconds for a final decision to be made by the individual interpreting the ECG. After I implemented this data into google sheet and calculated the average time for a human to interpret an ECG to be 36.4 seconds I was able to determine the data for the AI.

I implemented 15 different ECGs into AI for 3 trials. Each trial included the same 15 ECGs in the same exact order. I timed each ECG separately only starting the timer when AI began interpreting the ECG and ending it when AI provided me an answer. All the following data was recorded in google sheets and averaged in google sheets. In trial one, the average speed of AI came out to be 86.5 seconds, in trial two, the average speed of AI came out to be 85 seconds, in trial three, the average speed of AI came out to be 85.8 seconds. Once this data was collected, I took the overall average of the three trials which was estimated to be 85.8 seconds.

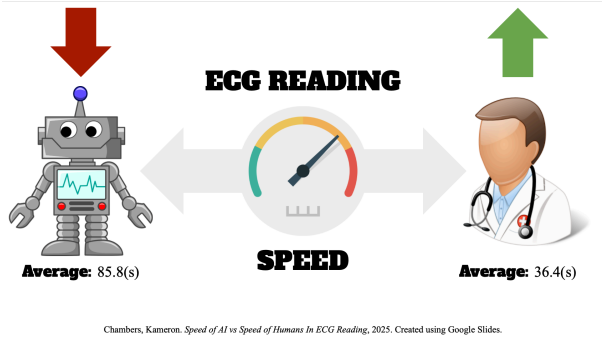
The data collected for both the humans and the AI were then compared. With humans having an average speed of 36.4 seconds and AI having an average speed of 85.8 seconds, it is visibly clear that humans have the general advantage in speed when it comes to providing an answer for whether or not an ECG reading is normal or abnormal. With cardiac complications happening at any moment in time, it's important that the one in charge of monitoring the ECG chart is able to determine when the ECG reading is abnormal. The quicker this information is determined, the quicker the patient has access to the care, treatments and test necessary to get them exactly what they need. Speed is a factor that is necessary to consider when determining whether or no AI should play a greater role in cardiovascular procedures. With AI being

approximately 49.4 seconds slower than humans, its fair to consider that AI is less efficient at producing quick answers.



Graph:

Chambers, Kameron. *Speed of AI vs Speed of Humans In ECG Reading*, 2025. Created using Google Sheets.



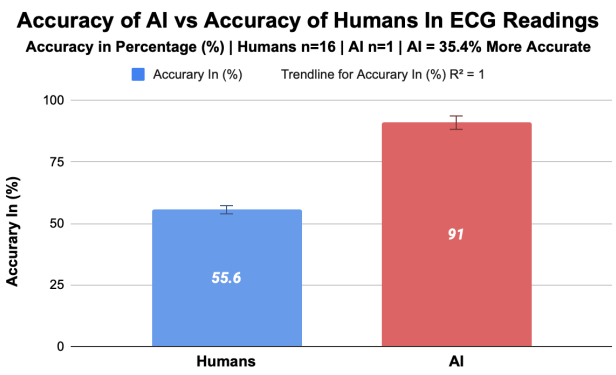
Chambers, Kameron. *Speed of AI vs Speed of Humans In ECG Reading*, 2025. Created using Google Slides.

For the second portion of this study AI provided more accurate answers than humans when it came to determining complication within ECG readings. The following information was collected from a publicly available research study conducted by Tahri Sqalli et al. (2021) which used 16 medical students in order to analyze 12-lead ECGs and collect data regarding speed, accuracy and eye-movement/eye-tracking. In this study, it was determined that the "average percentage of correct interpretations" appeared to be approximately 55.6% for the humans who participated in this study.

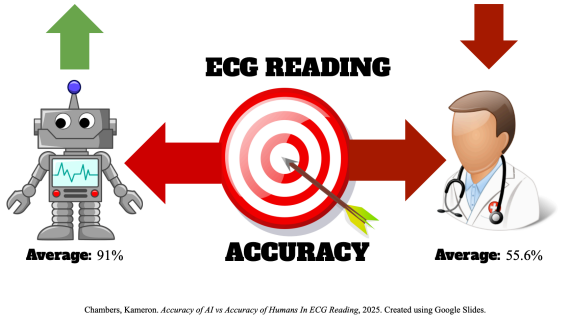
After comparing the answers provided by AI for all 15 ECGs in each of the 3 trials, I was able to gather the accuracy of AI in each trial and complete this data by taking the overall average. In trial one, AI got 3 ECG readings incorrect: ECG number 11, ECG number 14, ECG number 15. The score AI received in trial one was a 12/15 which is an 80% on this trial. In trial two, AI got 1 ECG reading incorrect: ECG number 13. The score AI received in trial two was

14/15 which is approximately a 94%. On the third and final trial, AI go 0 ECG reading incorrect, receiving a score of 15/15 which is 100%. I then used the final percentages (80%, 94% and 100%) to find the average accuracy of AI which came out to be approximately 91%.

The data for both the human and the AI were then analyzed and compared. With humans having an average accuracy of 55.6% and AI having an average accuracy of 91%, it is visibly clear that AI has the advantage in executing the correct answer. When it comes to the medical field, having a correct analysis and correct answer is crucial as it can greatly impact the lives of patients. Wrong interpretations and analysis can lead to unnecessary treatments and procedures which put the patient at risk for a false-interpreptation and reasoning. Accuracy is a more than necessary factor when it comes to the medical field and through this data and with AI being 35.4% more accurate than humans, it's fair to consider that AI is more accurate when it comes to analyzing ECGs.



Graph: Chambers, Kameron. *Accuracy of AI vs Accuracy of Humans In ECG Reading*, 2025. Created using Google Sheets.



Chambers, Kameron. *Accuracy of AI vs Accuracy of Humans In ECG Reading*, 2025. Created using Google Slides.

Analysis

In a side by side comparison, we see that both AI and humans have their ups, downs, pros and cons. In the medical field there is not much room for error and it requires everything to be done in a timely manner. Many situations, especially dealing with the cardiovascular system, can be life or death so both accuracy and speed are factors necessary to effectively provide the best care to patients.

With humans having the absolute advantage in speed and AI having the absolute advantage in accuracy, it's a fair split between the two. To answer the overall research question: "Does an AI model, trained to detect cardiac abnormalities in a ECG, perform with greater accuracy and speed than a human trained to read ECGs?" the answer is that AI performs with greater accuracy but not greater speed than humans.

The takeaway from this experiment that should be considered is that AI is something that doesn't have to be perceived or treated as an enemy rather as a tool and partner. As an example, while humans can flag ECGs for abnormalities, AI could double check and make sure the human notice is accurate. So should AI play a greater role in cardiac surgery than humans? The short answer determined by this data is no. There is still much room for improvement in AI and implementing AI into the medical field is an idea that should be considered in all medical facilities. As of right now and likely in the future, there will always be the need for humans to control and review the works of AI due to the fact its not perfect, just as humans. With AI also being a human creation, error is possible at any time.

AI should be incorporated into medical facilities in order to help with speed and accuracy but AI should not be prioritized over humans or have a greater role in the medical field than humans as there is much room for AI to improve, adapt and be created.

Procedure Errors & Complications

The first possible error in this study is the small sample size. With a human sample size of n=16 and an AI sample size of n=1, there is not enough participants for the data to be generalizable towards the larger population. In addition to this, the humans being specifically medical students is not representative of all individuals involved in the medical field as there are many doctors, physicians, etc; in this field of work. A solution to this problem is performing this research with a larger sample size or combining the elements of multiple research studies in order to create a research study with more generalizable data.

The second possible error in this study is the insufficient materials towards creating an AI program eligible to read ECGs. With the AI used in this study being created through a basic code and on a MacBook, there were many limitations set on this AI. Some limitations included only being able to read a certain number of ECGs before crashing and only being able to be trained to read a limited number of ECGs. A solution to this problem is to use more sufficient resources in a controlled environment to create an AI ECG analyzing program that more accurately reflects the AI ECG analyzer that would be incorporated into the medical field.

The third and final possible error in this study is the limitations towards data. With the lack of research regarding this topic, there was a limit towards the data that was eligible for public usage. The possible solution to this issue is performing more research regarding this topic as it can help towards determining the future of employment and how AI will impact employment. This is also a rising question and concern so research on this topic is a great idea.

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